Teaching Economics to the Machines

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Black–Scholes Model Overview

Motivation

The Transfer Learning Framework

An application in option pricing

Conclusion

Model Dynamics

We assume the underlying asset price S_t follows a *geometric Brownian motion*:

$$dS_t = \mu\, S_t\, dt \;+\; \sigma\, S_t\, dW_t,$$

- μ: drift (annualized expected return)
- σ: volatility (annualized stdev of returns)
- W_t: standard Brownian motion

Under the *risk–neutral* measure, replace μ with the risk-free rate r.

Black–Scholes Pricing Formulas

For a European *call* and *put* option:

$$C = S_0\,\Phi(d_1) \;-\; Ke^{-rT}\,\Phi(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)\,T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

$\Phi(\cdot)$ is the standard normal CDF.

Key Parameters

Key inputs: C, S_0, K, T, r, σ.

Symbol	Name	Description
C	Call option price	Theoretical price of a European call option.
S_0	Spot price	Current price of the underlying asset.
K	Strike price	Exercise price of the option.
T	Time to maturity	Time until expiration (in years).
r	Risk-free rate	Continuously compounded annual rate.
σ	Volatility	Annualized stdev of log-returns.

Black–Scholes Assumptions

Market Structure:

- No arbitrage and complete markets (perfect replication)
- Continuous trading and frictionless markets

Underlying Dynamics:

- Geometric Brownian Motion
- Constant interest rate r and volatility σ

Contract Specification:

- European option
- No dividends (or constant dividend yield q)

In practice?

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The End of Theory?

Chris Anderson, Wired (2008):

"All models are wrong, but some are useful." So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? ... Today companies like Google, which have grown up in an era of massively abundant data, don't have to settle for wrong models. Indeed, they don't have to settle for models at all.

Ilya Sutskever, Reuters (2024):

The 2010s were the age of scaling, now we're back in the age of wonder and discovery once again ... Scaling the right thing matters more now than ever.

How to teach a robot to play pool?

How should we teach a robot to play pool?

- **Data-driven approach:**
  - x: shot angle, position, force ...
  - y: ball direction, speed, spin ...
  - However, labeled data are often expensive to collect.
- **Theory-driven approach:**
  - Model of elastic collisions
  - Conservation of linear momentum and kinetic energy $\Rightarrow$ complete predictions for ball movements
  - However, the model is likely misspecified (inelastic collision, imperfect surface ...)
- **Why not combine the knowledge from theory and data?**
  - Our answer: transfer learning

Overview of results

- Transfer learning (TL): First learn from theory (through synthetic data), then from real data (with fine-tuning).
- In an application to option pricing, TL significantly outperforms a conventional deep learning (DL) model and a popular structural model.
- This is especially true:
  - for median and long time-to-maturity, OTM, liquid options;
  - when the market condition is volatile; or
  - when the input feature appears "uncommon" relative to the training sample.
- TL is more stable than DL. Its advantage over DL is more pronounced in smaller samples.
- TL can help identify features missing from the structural model.

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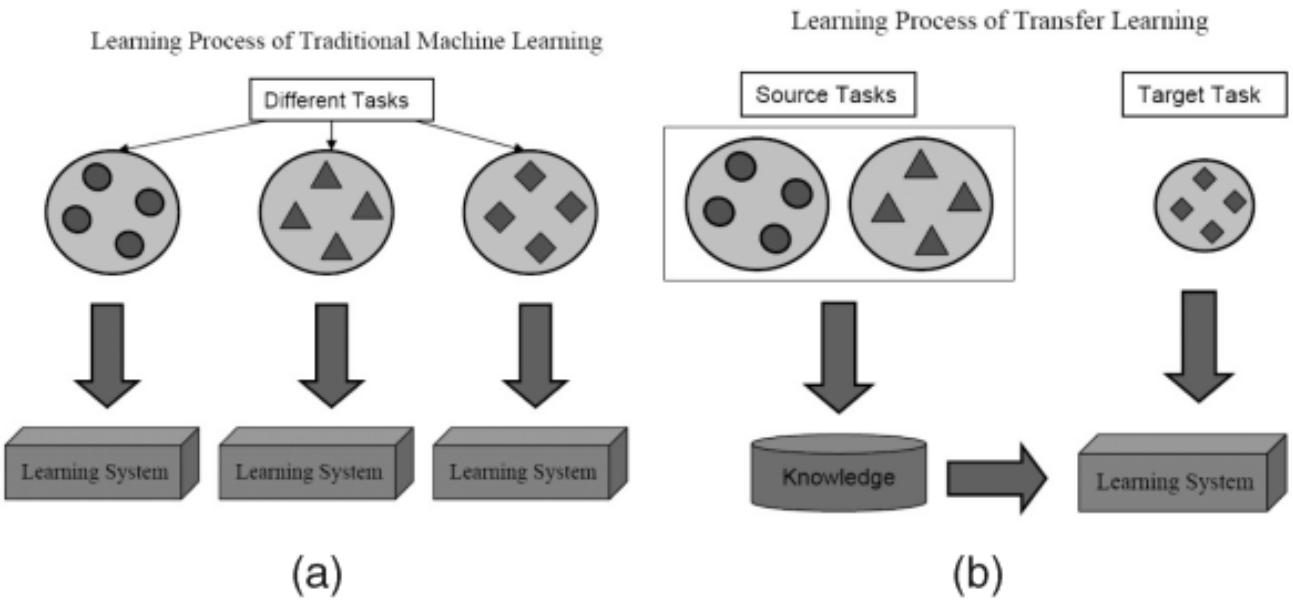
Transfer learning

- **Transfer Learning:** Using pre-trained models on large datasets (source domain) to accelerate learning on a smaller dataset (target domain) for a specific task.
  - computer vision: pre-train CNNs on ImageNet $\Rightarrow$ medical image analysis
  - natural language processing: pre-trained LLMs $\Rightarrow$ sentiment analysis
- **In our setting:** Source domain = economic model; target domain = empirical data.



Teaching economics to the machines

- The neural networks trained on the simulated data from a theoretical model inherit the structural restrictions from the theory.
- TL: Use theory-implied parameter values as the starting point. Fine-tuning allows the structural restrictions to regulate the learning on real data.
- **Why might this be helpful?**
  - Bias-variance tradeoff: Valid restrictions help with learning. Misspecified models also add bias.
  - Theory-informed initialization can mitigate issues in numerical optimization, such as avoiding local minima and saddle points.
- **Potential benefits:**
  - Variance reduction
  - Speed up training on real data; less demand for real data
  - Improve generalizability beyond training data boundaries

The setup

- Denote by $\mathcal{X}$ and $\mathcal{Y}$ the input and target space, respectively, with unknown probability distribution $\mathbb{P}(\mathcal{X}, \mathcal{Y})$.
- Denote a neural network with L layers as

$$F(L; \sigma_1, \cdots, \sigma_L; W_1, \cdots, W_L).$$

  - σ_i: activation function
  - $W = [W_1, \cdots, W_L]$: network weights
- Standard approach to train a neural network: Find a training set $S = \{(x_i, y_i)\}_{i=1}^m$ and a loss function $\mathcal{L}(f(x), y)$. Then minimize the empirical loss:

$$W^* = \arg\min_W \frac{1}{m} \sum_{i=1}^m \mathcal{L}(F(x_i; W), y_i).$$

The setup

- Economic model: Could lead to a fully specified joint distribution for x and y as $\mathbb{Q}(\mathcal{X}, \mathcal{Y})$. Or, it could just be the conditional expectation of the target y,

$$\mathbb{E}^{\mathbb{Q}}[y|x] = g(x)$$

- A model is typically designed to be parsimonious, with a small number of states, some of which can be latent. There are also model parameters.
- The model can be misspecified: $\mathbb{Q} \neq \mathbb{P}$.

Source domain

- Generate synthetic data $((\tilde{x}_i, \tilde{y}_i))_{i=1}^{M}$ from the model-implied distribution, $\mathbb{Q}(\mathcal{X}, \mathcal{Y})$.

$$\tilde{y}_i = g(\tilde{x}_i)$$

- In the source domain, we train the neural networks to learn from the synthetic dataset:

$$\widetilde{W}^* = \arg\min_{\widetilde{W}} \left(\lambda_0 \mathcal{L}_0 + \sum_{j=1}^{K} \lambda_j \mathcal{L}_j \right)$$

- Economics-informed losses:

$$\mathcal{L}_0 = \sum_{i=1}^{M} w_{0i} \left| F(\tilde{x}_i | \widetilde{W}) - \tilde{y}_i \right|$$

$$\mathcal{L}_j = \sum_{i=1}^{M} w_{ji} \left| \partial_j F(\tilde{x}_i | \widetilde{W}) - \partial_j g(\tilde{x}_i) \right|, \quad \forall j > 0$$

$\hookrightarrow$ Physics-informed losses: Raissi et al. [2017]

Target domain

- In the target domain, train with real data $\big((x_i, y_i)\big)_{i=1}^{n}$.
- Identical network architecture as in the source domain; use the weights from the source domain for initialization:

$$\widehat{W}^* = \arg\min_{\widehat{W}} \sum_{i=1}^{m} w_i \big| F(x_i | \widehat{W}) - y_i \big|,$$

  with initial weights $\widehat{W}^*$.
- Fine-tuning: Small learning rate and early stopping.
- Alternative approaches:

  $\hookrightarrow$ Augment the network from the source domain with new layers.

  $\hookrightarrow$ Partial fine-tuning: fixing bottom part of the network (frozen layers).

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Example application

- A horse race between theory-based transfer learning (TL) and standard deep neural networks (DL) for option pricing.

  $\hookrightarrow$ Use the Black-Scholes model in the source domain.
  $\hookrightarrow$ Augment the set of features with past returns, bid-ask spreads, put-call ratios ...(total of $d = 16$ features)

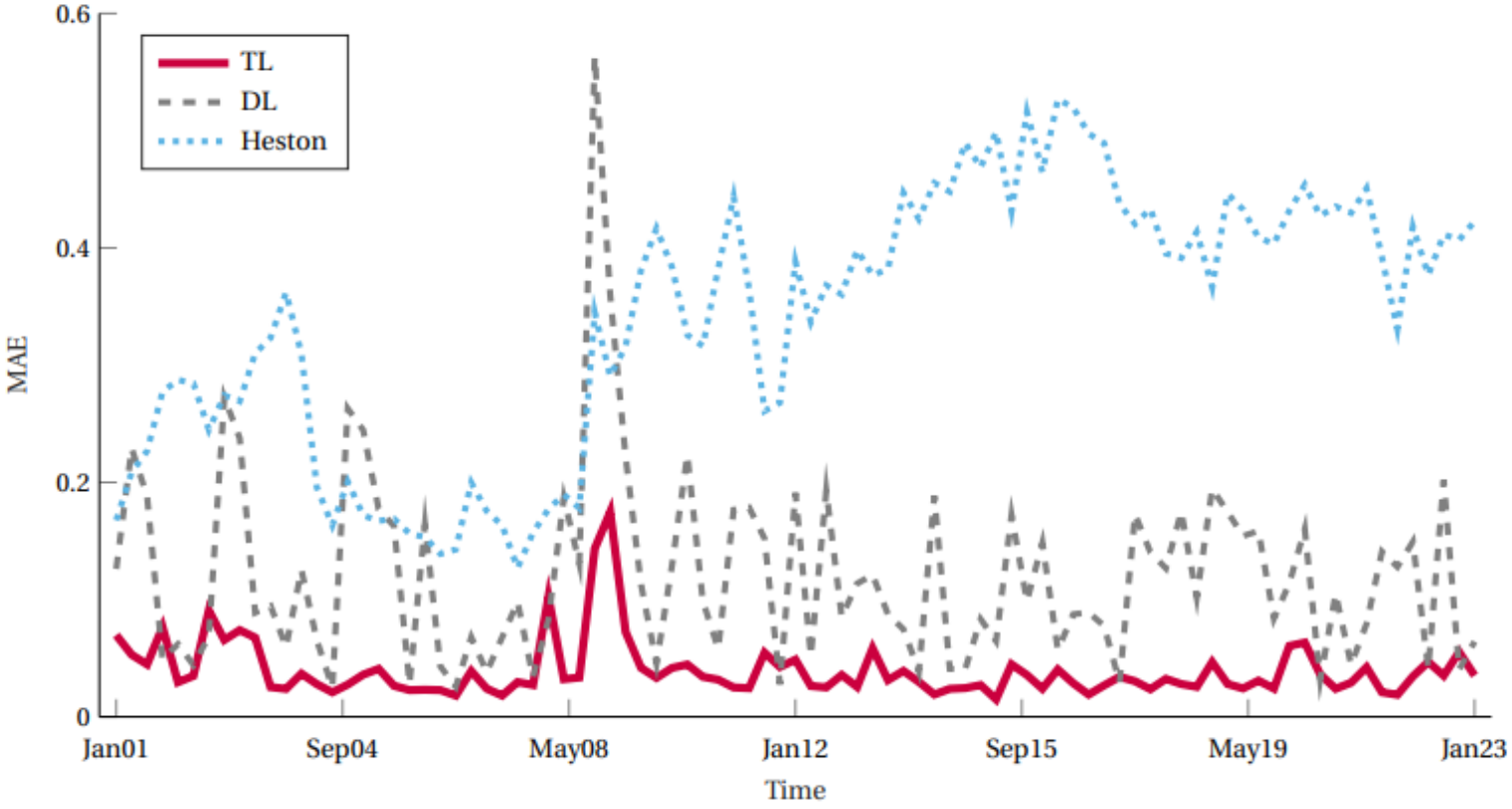
- Architecture: 16 hidden-layer ResNet, 22 neurons in each layer ($> 7,800$ trainable parameters), LeakyRelu activation

- Rolling window: At the end of each quarter, use data from the past 3 months to train the models; use the models to predict option prices in the next 3 months.

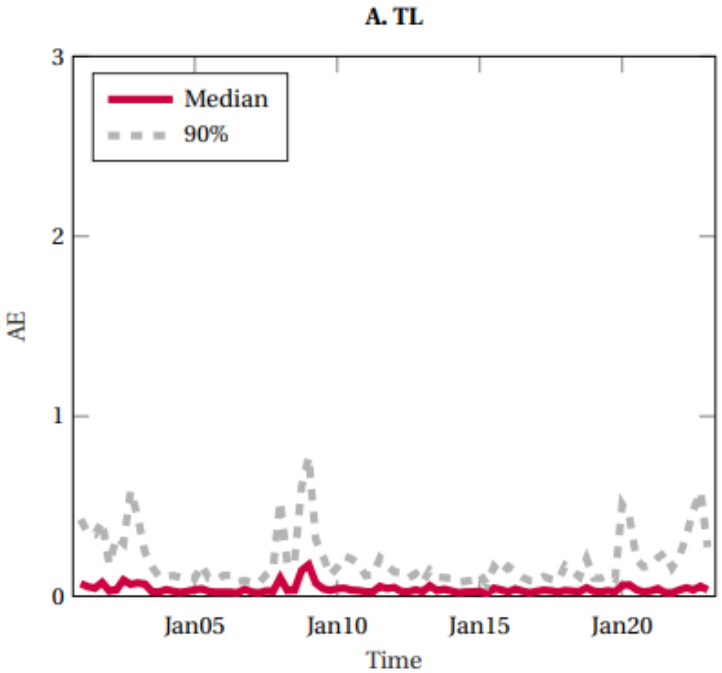
  $\hookrightarrow$ Mimicking a setting with limited labeled data in the target domain.

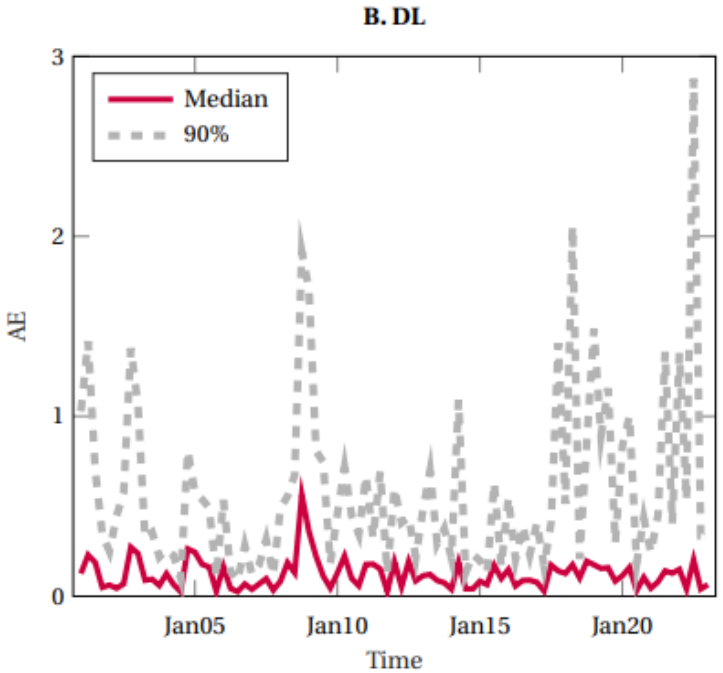
- Other candidate models: Heston model, DL with BS penalty, boosted BS

TL outperforms DL and Heston out of sample (based on IV-MAE)



TL has less extreme outliers

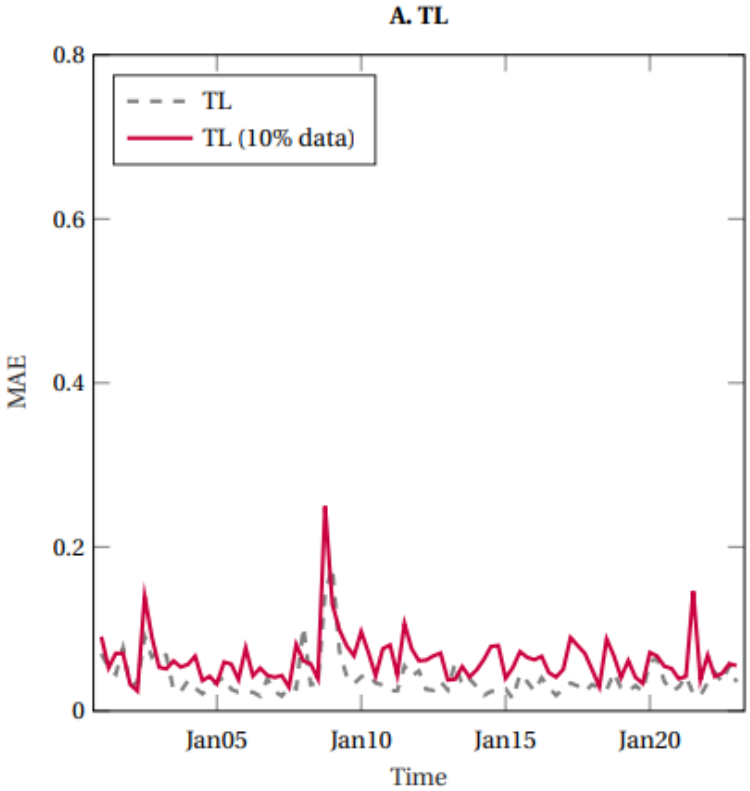


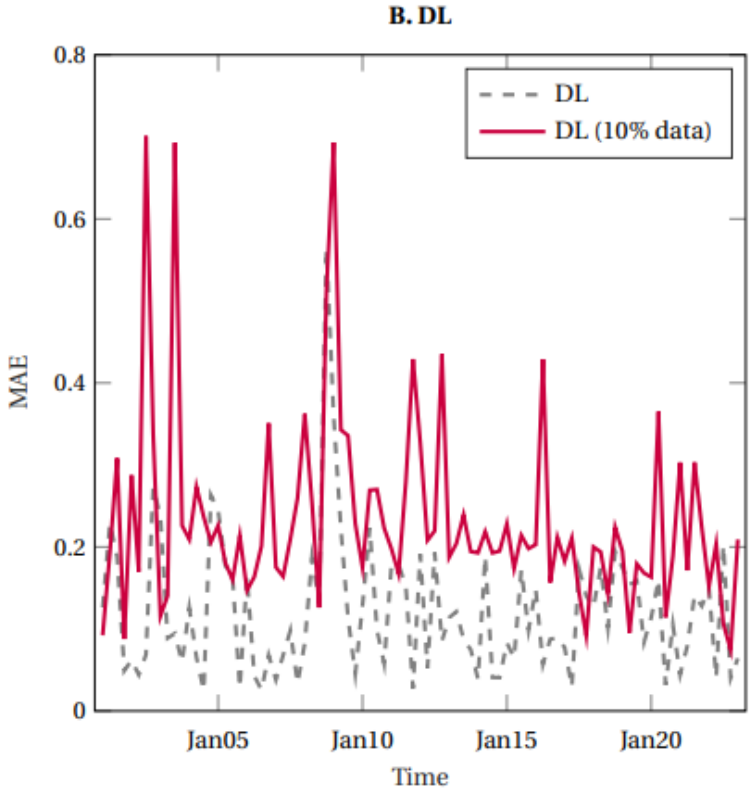


Performance attribution

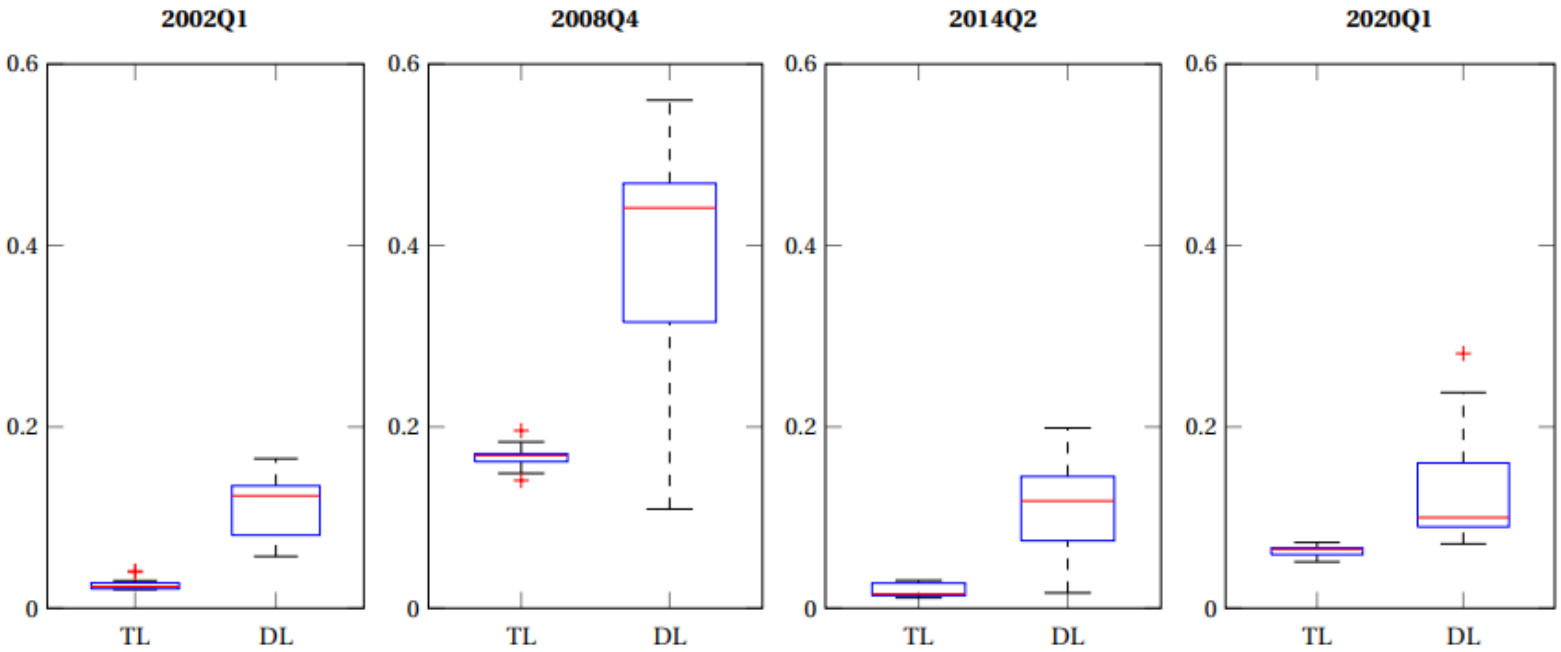
- Regressing the difference in absolute pricing errors between the DL and TL model at contract-day level on a variety of attributes.
- TL has a bigger advantage relative to DL:
  - for options with median and long maturity (>14 days), OTM (especially DOTM), with better liquidity (high volume);
  - when the market is more volatile;
  - when volatility spikes;
  - when *test sample* is "uncommon" relative to training sample (Mahalanobis distance).

TL is better at learning from a smaller sample





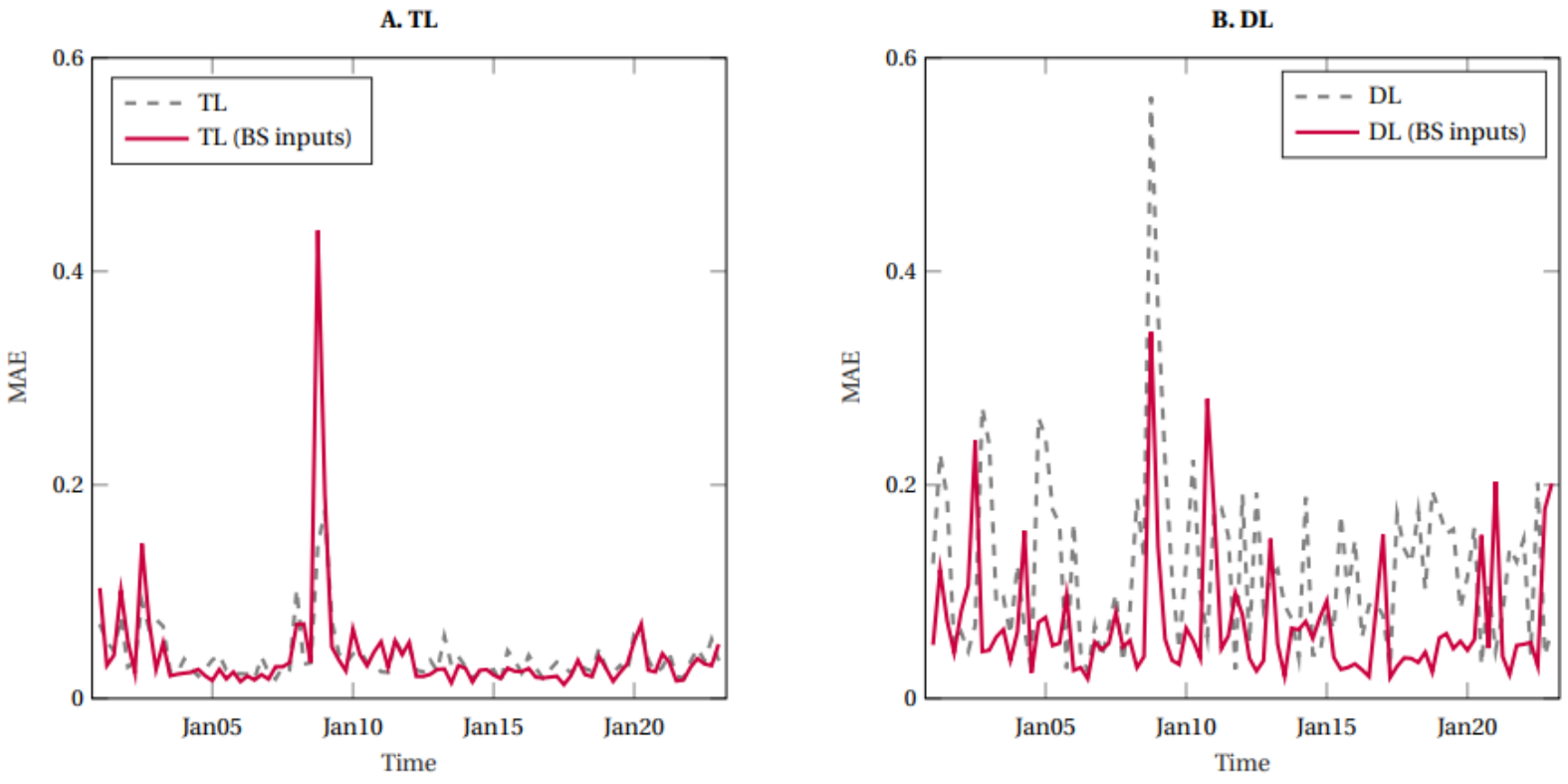
TL is less sensitive to the randomness of training



What is source of TL out-performance?

- **Theory helps identify relevant inputs?**
  $\hookrightarrow$ Compare to DL with the same inputs
- **More data?**
  $\hookrightarrow$ Rolling-window vs. expanding-window
  $\hookrightarrow$ Pooling real data with simulated data
- **Other ways to bring in structural model information**
  $\hookrightarrow$ Introducing a penalty in the loss function based on the violation of BS restrictions.
  $\hookrightarrow$ Boosting BS with OLS.
- **Would any restrictions work?**
  $\hookrightarrow$ Use no-arbitrage restrictions in the source domain

Theory does more than identifying inputs



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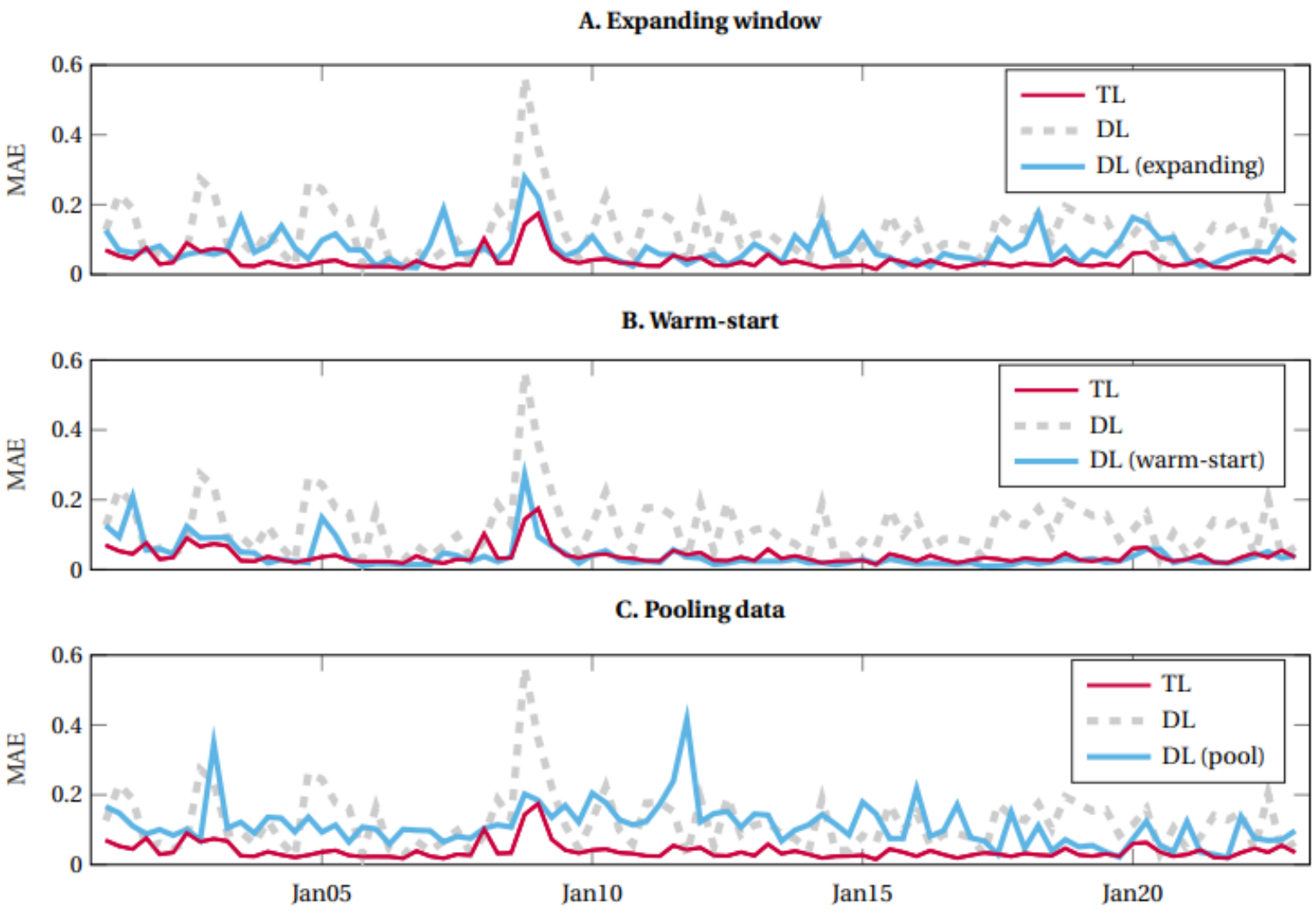
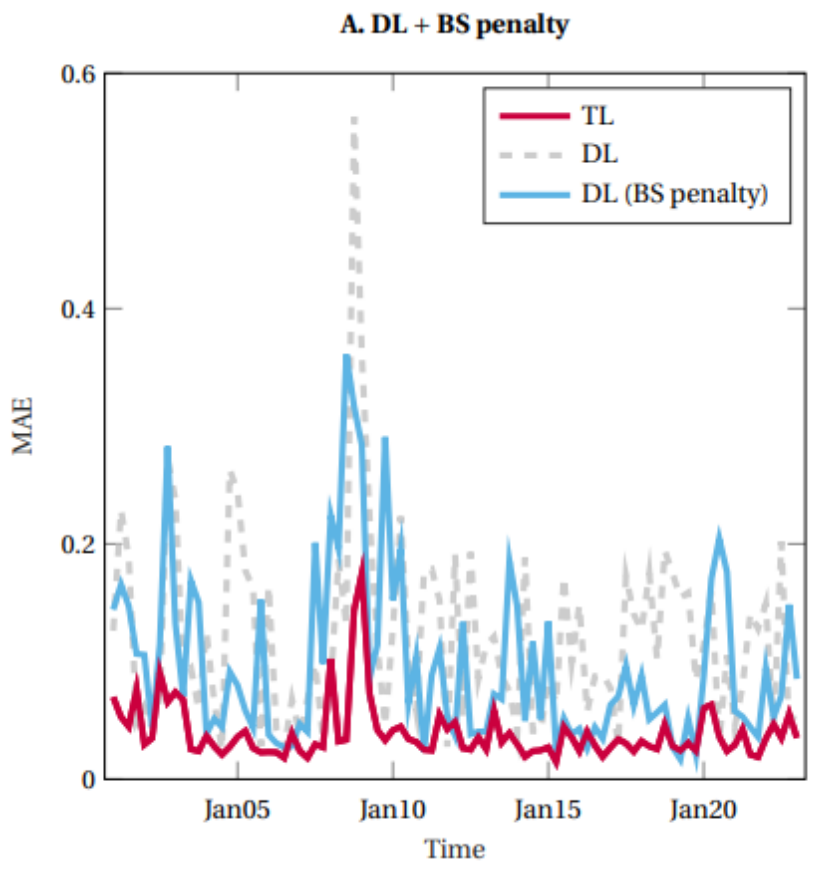
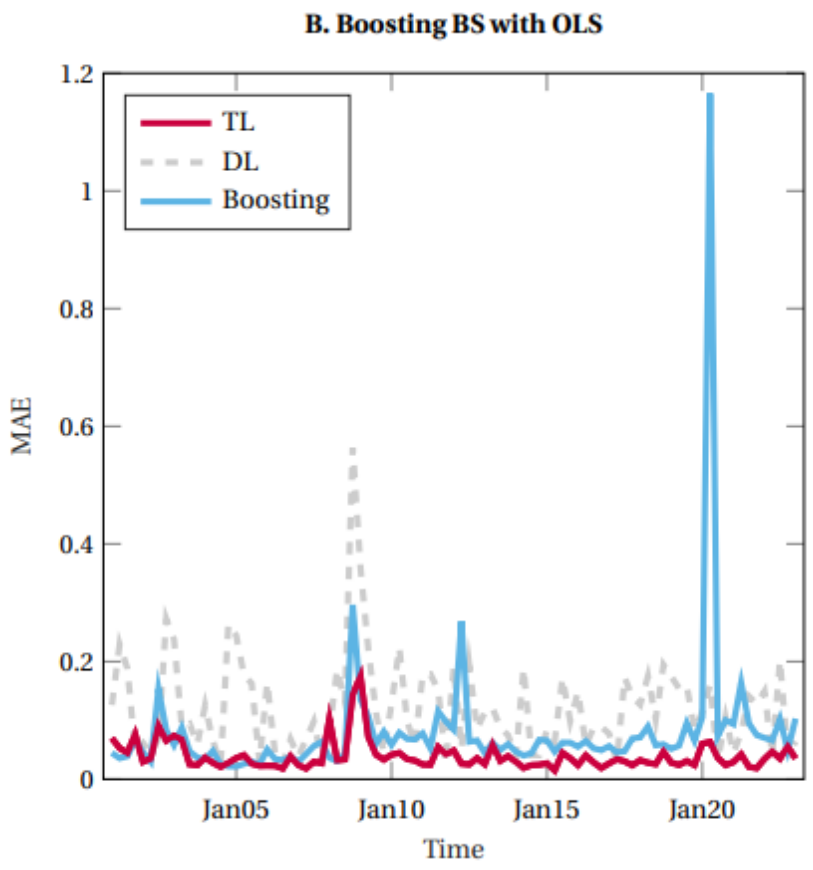
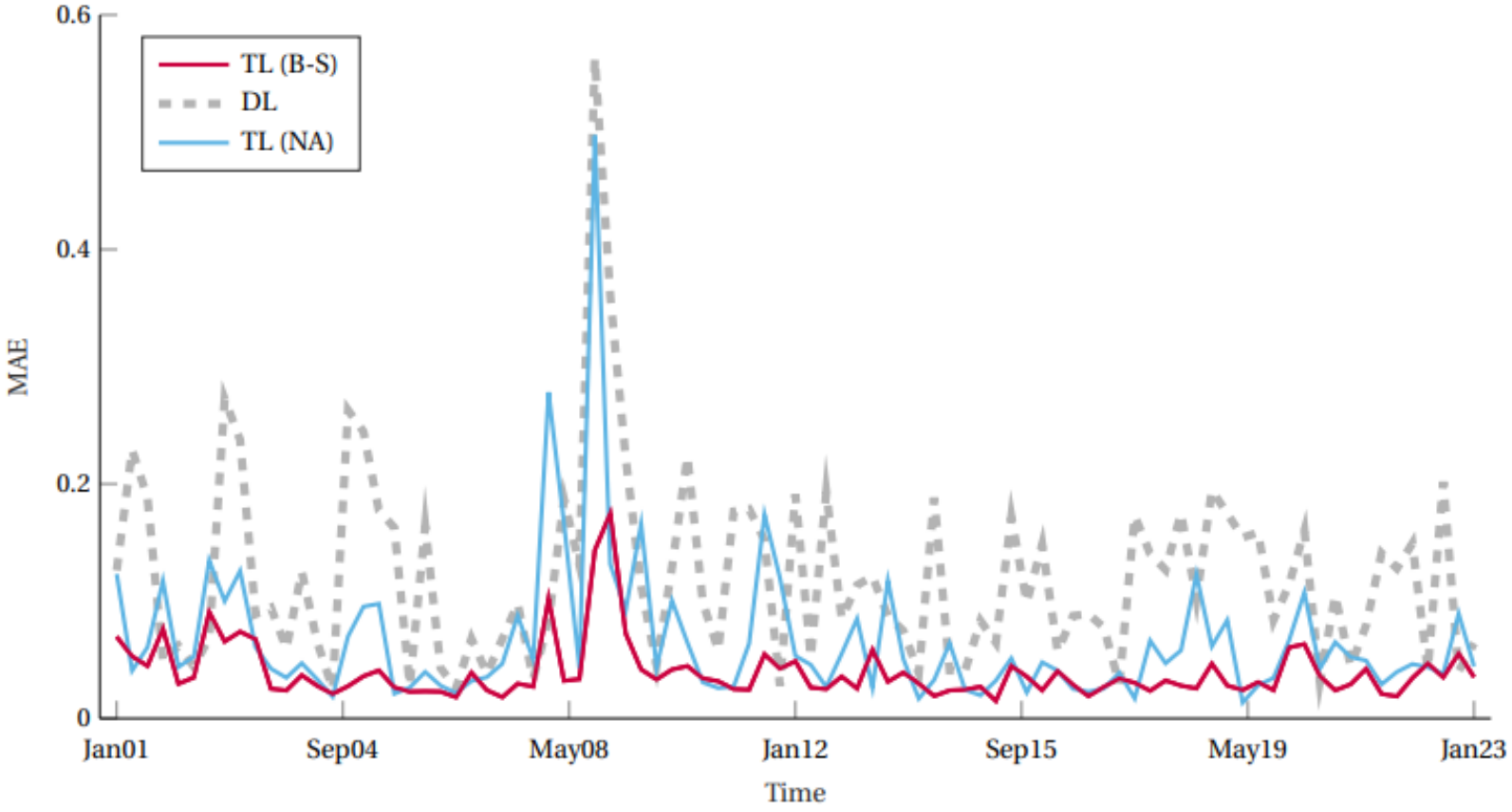


Alternative ways to incorporate model information

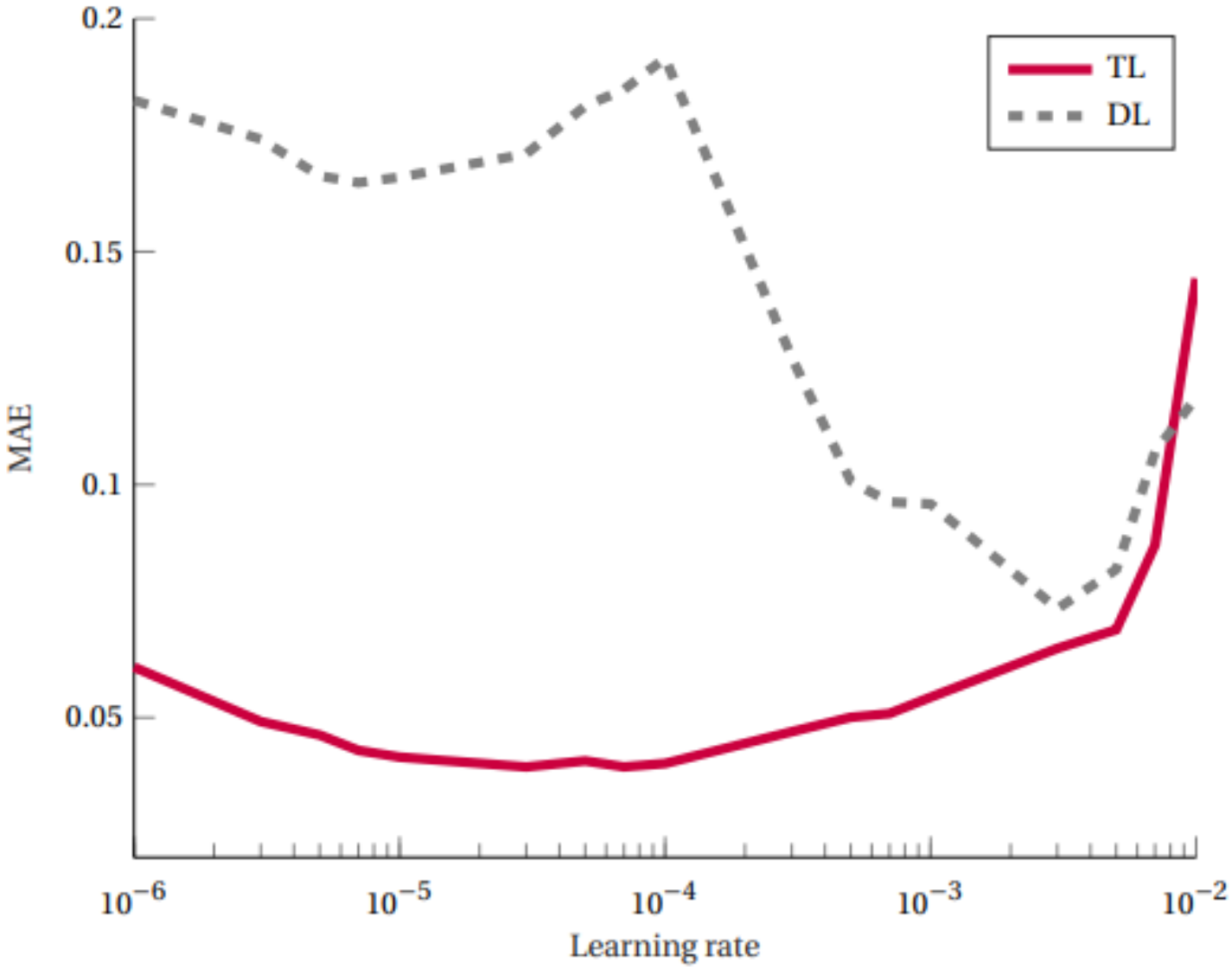




Using no-arbitrage restrictions in the source domain



The importance of fine-tuning



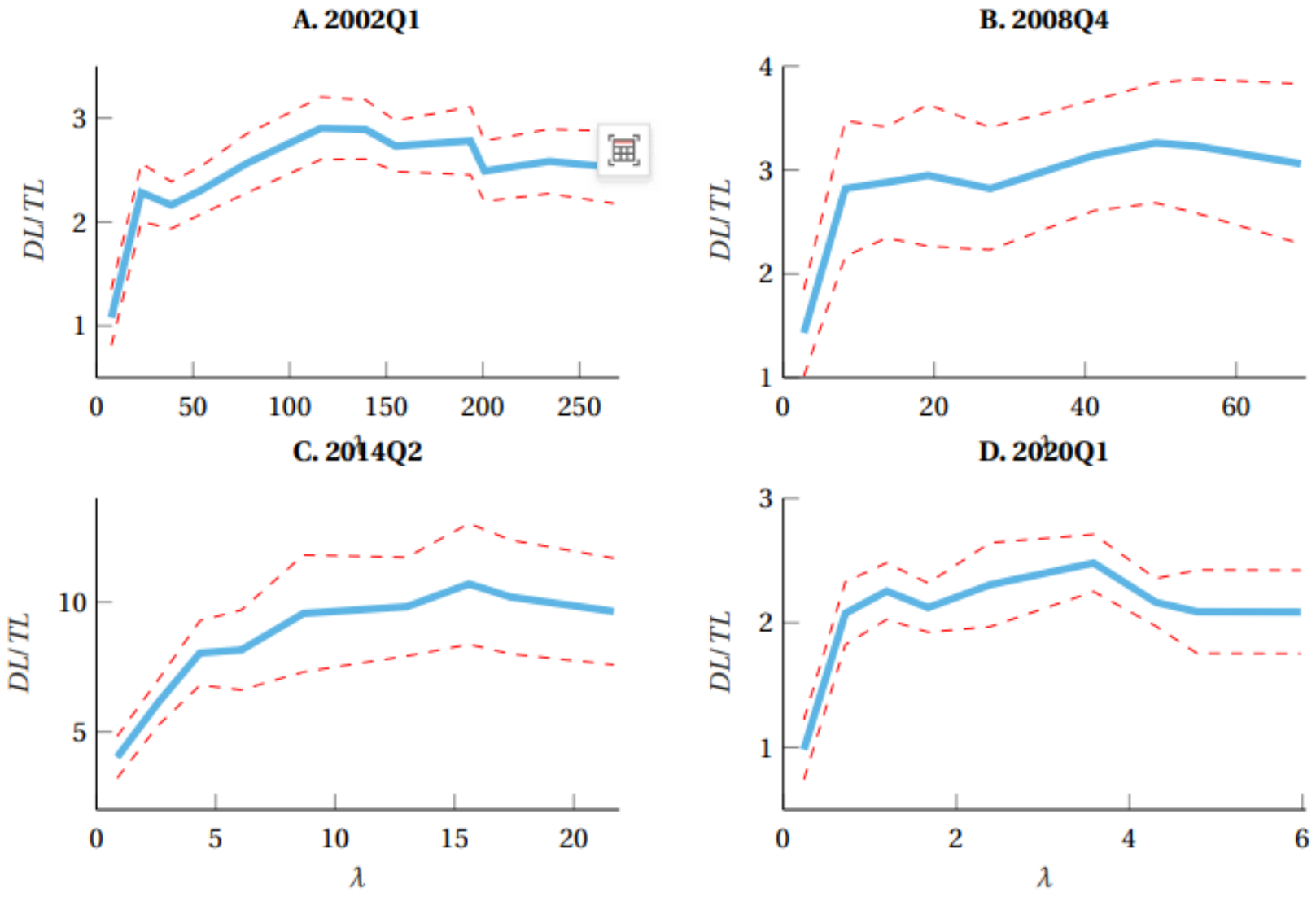
TL $\neq$ Informative Prior

- Consider relative sample size

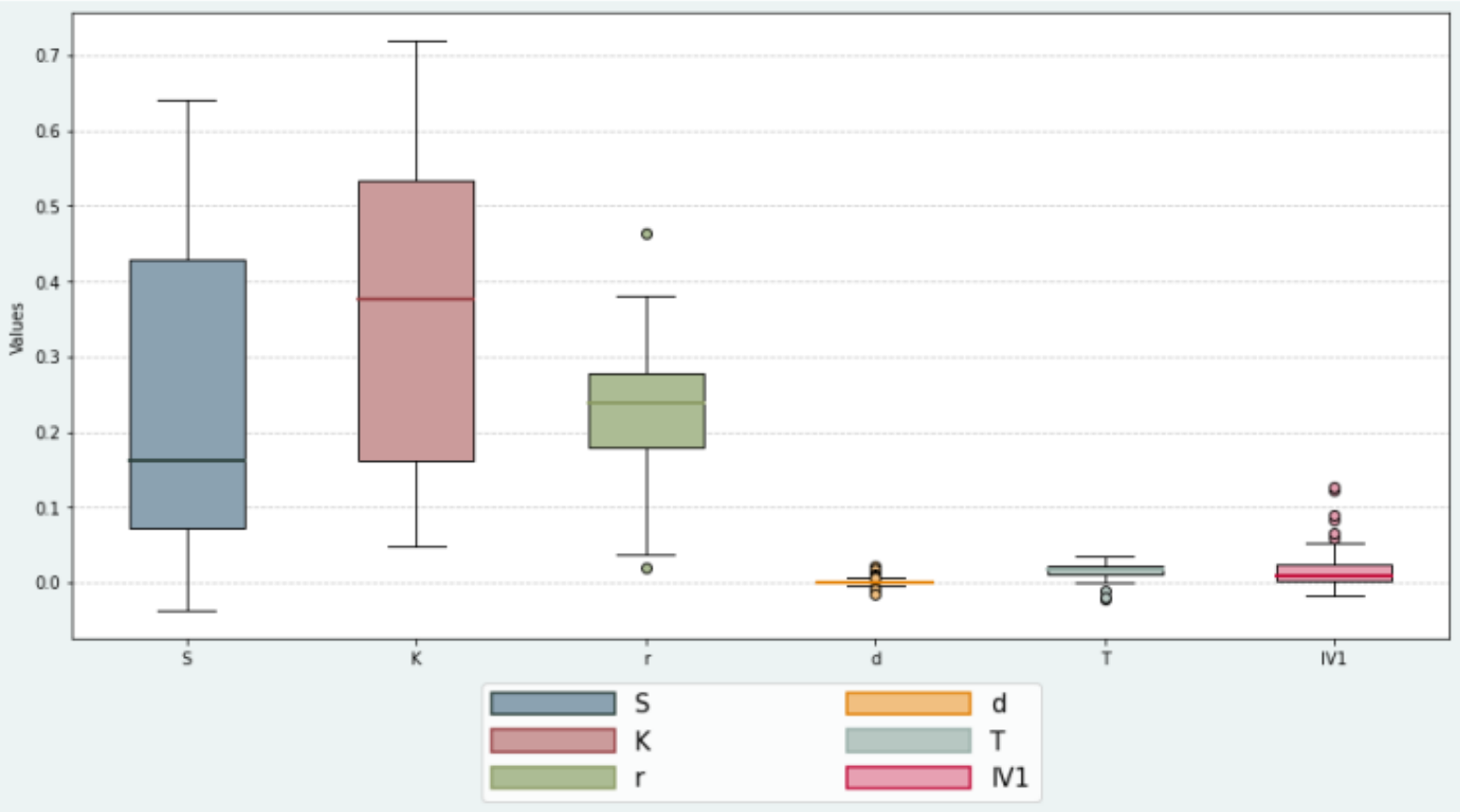
$$\lambda = \frac{\text{sample size of synthetic data}}{\text{sample size of real data}}$$

- In Bayesian VAR, λ effectively controls the prior precision.
- Del Negro and Schorfheide [2004]: An inverse U-shaped relation between forecasting accuracy and λ.
  - Intuition: Too little or too much weight on the DSGE model can both hurt performance.
- TL: More synthetic data will improve the accuracy of the structural model. But the learned weights in the source domain are still only the initial values for the next stage.
- How much the final weights will deviate depends on:
  - Gap between the structural model and true DGP
  - Hyperparameters of learning in the target domain: learning rate, epochs

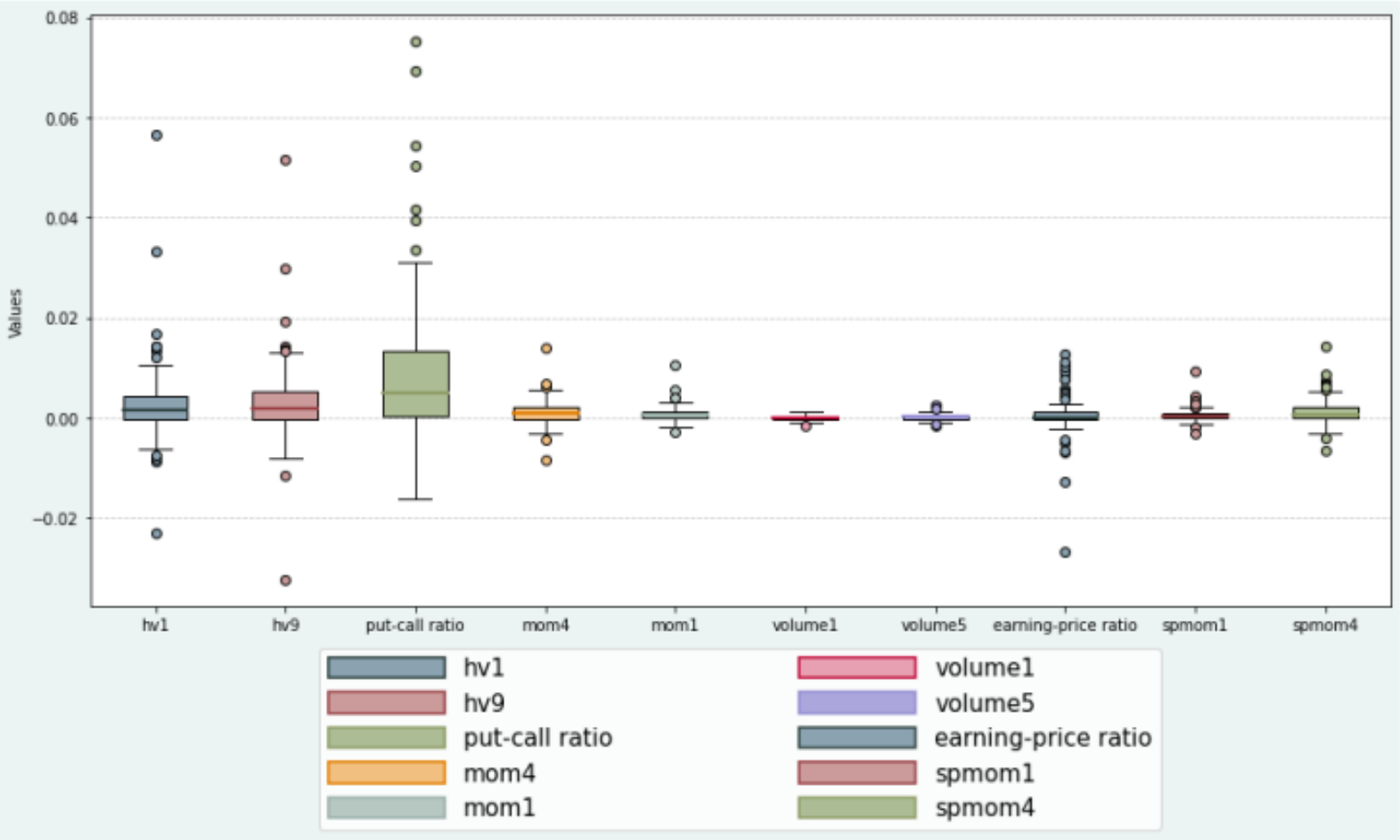
TL is better at learning from a smaller sample



Feature importance



Feature importance



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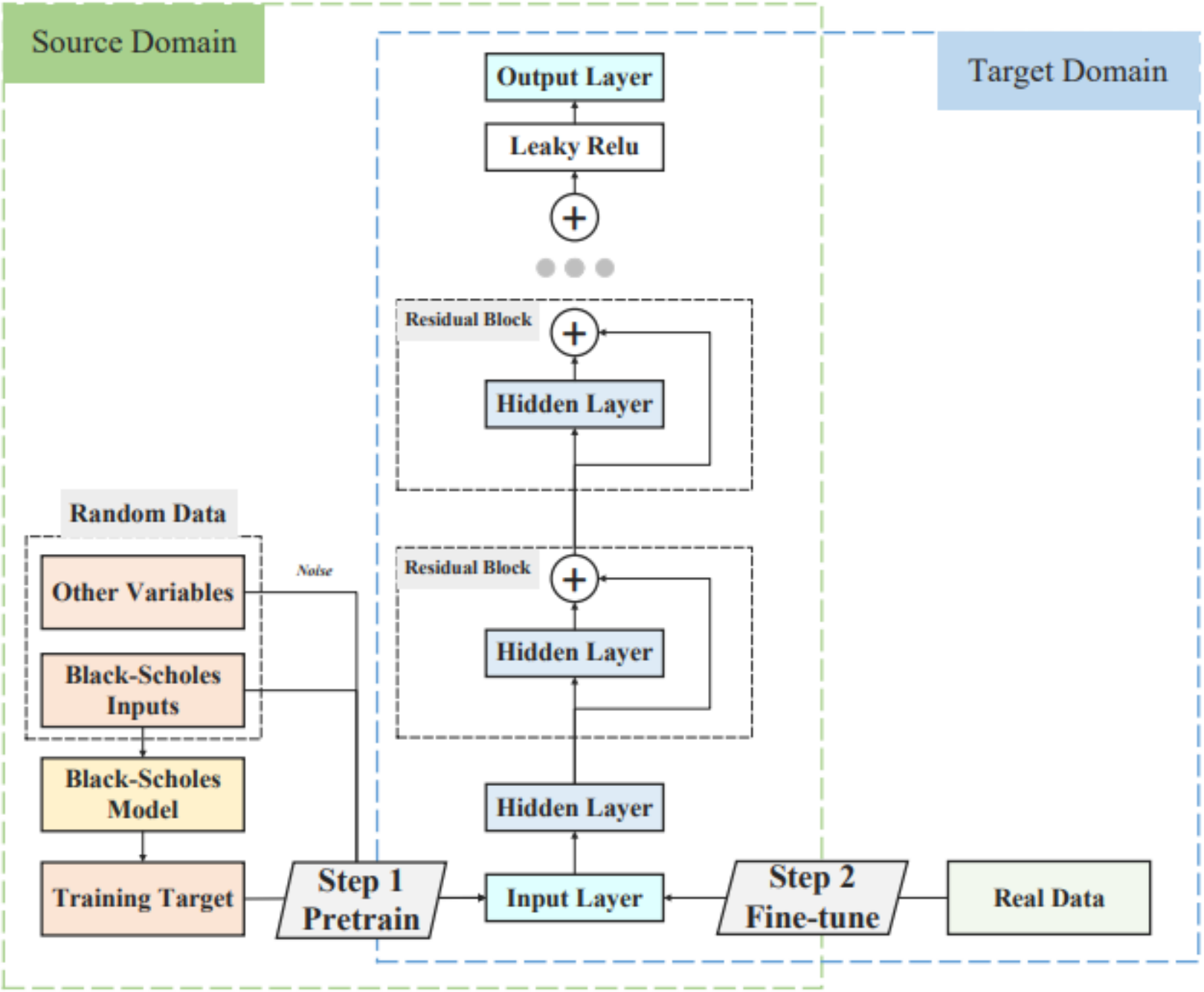
- **Guiding ML with economic theory through transfer learning.**
  - Higher accuracy and stability, especially in smaller samples.
  - Generalizability. Dealing with structural breaks.
- **Not specific to neural networks. The framework easily applies to other ML models.**
  - More useful when real data are scarce + but still need flexibility
- **Work in progress:**
  - macro forecasting
  - prepayment model
  - credit risk model
  - portfolio construction
  - ...

Theories are dead? Long live theories!


Source Domain
Target Domain
Output Layer
Leaky Relu
Residual Block
Hidden Layer
Residual Block
Hidden Layer
Hidden Layer
Input Layer
Random Data
Other Variables
Black-Scholes Inputs
Black-Scholes Model
Training Target
Noise
Step 1 Pretrain
Step 2 Fine-tune
Real Data

Marco Del Negro and Frank Schorfheide. Priors from general equilibrium models for vars. *International Economic Review*, 45(2):643–673, 2004.

Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics informed deep learning (part i): Data-driven solutions of nonlinear partial differential equations. *arXiv preprint arXiv:1711.10561*, 2017.